

Home assignment: day 4 → day 7

Deadline: 2026-07-07, 14:59.

1. Solve the recurrence equation:

$$a_n + 6a_{n-1} + 9a_{n-2} = 0, \quad a_0 = 5, a_1 = 10.$$

2. Solve the recurrence equation:

$$c_n + 6c_{n-1} + 10c_{n-2} = 0, \quad c_0 = 0, c_1 = 10.$$

3. Solve the recurrence equation:

$$a_n = (n + 5)a_{n-1}, \quad a_1 = 100.$$

You may certainly use factorial in your answer!

4. I enter a casino in Las Vegas with 50 tokens. I play the roulette. I always bet one token on black. I will leave the casino when my fortune will be 70 tokens or 0 tokens.

In a typical Las Vegas roulette there are 38 pockets: 18 black pockets, 18 red pockets, one zero and one double zero. Let p_n be the probability that I will leave the casino as winner if I have n tokens now.

- What are the initial conditions p_0 and p_{70} ?
- Build the recurrence equation for p_n .
- Solve the recurrence equation for p_n .
- Evaluate p_{50} up to 2 digits.

Demo for the quiz on day 5

1. Consider the recurrence equation $y_n - 7y_{n-1} + 10y_{n-2} = 0$.

Find all solutions of the form $y_n = \lambda^n$.

2. You know that $r_n = c \cdot 2^n + d \cdot n \cdot 2^n + 4$ is the solution of some recurrence equation with initial conditions $r_0 = 6$ and $r_1 = 10$.

Find c and d .
